

University of Rwanda/CST

Assignment of MATH I

November 11, 2024

1. (a) Use the formal definition to prove that $\lim_{x \rightarrow 1} f(x) = 1$ if

$$f(x) = \begin{cases} x^2 & \text{if } x \neq 1 \\ 1 & \text{if } x = 1 \end{cases}$$

- (b) Suppose $|f(x)| \leq g(x)$ for all x . What can conclude about $\lim_{x \rightarrow a} f(x)$ if $\lim_{x \rightarrow a} g(x) = 0$? What if $\lim_{x \rightarrow a} g(x) = 3$?
- (c) Apply formal definition of the limit to prove that if f is continuous function at L and if $\lim_{x \rightarrow c} g(x) = L$, then $\lim_{x \rightarrow c} f(g(x)) = f(L)$.
2. Evaluate $\lim_{x \rightarrow 0} \frac{x^2 - 3x + 2}{x^3 - 2x}$ as
- (a) $x \rightarrow 0^+$
- (b) $x \rightarrow 2^-$
- (c) What, if anything, can be said about the limit as $x \rightarrow 0$.

3. Find an equation of the tangent to the curve $2x + y - \sqrt{x} \sin(xy) = \frac{\pi}{4}$ at $x = \frac{\pi}{4}$

4. Compute

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x} + \sqrt{1-x} - 2}{x^2}.$$

5. Evaluate $\lim_{x \rightarrow 3} f(x)$ if

$$f(x) = (x+2) \frac{\sqrt{x^2 - 6x + 9}}{x^2 - 7x + 12}.$$

6. Find equations of any lines that passes through the point $(-1, 0)$ and are tangent to the curve $y = \frac{x-1}{x+1}$.

7. Find y'' if $x^3y - 3xy + x^2y^3 = x$.

8. The current flowing in a particular circuit as a function of time t is given by $i = e^{3t} \sin t$. Determine the charge q which has passed through the circuit in $[0, t]$. Recall that $i = \frac{dq}{dt}$.

9. The root-mean-square value of a function $f(x)$ over an interval $[0, T]$ is given

$$f(x)_{rms} = \sqrt{\frac{1}{T} \int_0^T [f(x)]^2 dx}.$$

Find the root-mean-square value of the the following functions on the indicated interval

- (a) $f(x) = \sqrt{\sin^{-1} x}$ over $[0, 1]$.
 (b) $f(x) = \sqrt{e^x \cos x}$ over $[0, \frac{\pi}{2}]$.

10. Evaluate 2

- (a) $\int \frac{\sec^{-1} x}{x\sqrt{x^2 - 1}} dx.$
 (b) $\int x^3 \sin x^2 dx.$
 (c) $\int_0^\infty e^{-x} \sin 4x dx.$
 (d) $\int \frac{12x^4 + 190x^2 + 13x - 6}{(2x - 1)(x^2 + 16)} dx.$
 (e) $\int \sec^6 x \tan^8 x dx.$
 (f) $\int \frac{\cos^4 x}{\sin x} dx.$
 (g) $\int_0^1 \frac{1}{\sqrt{x(1-x)}} dx.$

11. Find the area between region enclosed by the curves $y = x^3$ and $y = |x|$.
 12. Find the volume of the solid generated when the region between the graphs of $f(x) = 12 + x^2$ and $g(x) = x$ over the interval $[0, 2]$ is revolved about the x -axis.

Date of submission: November, 18th 2024.